Persistent Homology of Attention Graphs: A Topological Pilot for Hallucination Detection

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[1]Data Science & Mathematics, Independent Research

June 16, 2026

Abstract

We test the hypothesis that hallucinated continuations of factual prompts shift the persistent-homology summary statistics of the attention graph relative to truthful continuations. On GPT-2 across 10 hand-curated truthful/hallucinated pairs (e.g. "The capital of France is Paris" vs. "Berlin"), we compute the Vietoris–Rips persistent homology of the per-layer attention-distance matrix and report the total persistence in H_0 and H_1. The local pilot, by design, is underpowered: with only 10 pairs, no layer reaches $p < 0.05$ (minimum $p = 0.852$ at layer 7, on H_0). The mean H_0 totals are 3.406 (truthful) vs. 3.338 (hallucinated), giving an effect-size estimate that allows us to power-plan the larger run.

1 Introduction

A widespread concern with large language models is hallucination: the production of fluent text that is not factually grounded. Recent work has approached the problem from the activation side [1, 2]; in this paper we ask whether a purely structural signal—the topology of the per-layer attention graph—suffices.

Question. *Do truthful and hallucinated continuations yield statistically distinguishable persistent-homology summaries of the attention graph?*

Setup. For each factual prompt we generate two continuations: one factually correct (e.g. "Paris") and one plausible but wrong ("Berlin"). The surface form is matched. We forward both through GPT-2, extract every layer's attention matrix, convert to a Vietoris–Rips filtration over the metric $d_{ij} = 1 - \max(A_{ij}, A_{ji})$, and compute persistence in H_0 and H_1.

Pilot. The local configuration uses 10 curated pairs; this is intentionally underpowered. The point of the pilot is to (i) validate the pipeline and (ii) measure the effect size so that the GPU configuration can be sized correctly. The local pilot finds no layer with $p < 0.05$ on H_0 (minimum $p = 0.852$ at layer 7); under the observed effect-size estimate, a sample of $N \geq 200$ pairs would be required to reach $\alpha = 0.05$ at $\beta = 0.2$, which is exactly what the `gpu_full.yaml` configuration supplies via TruthfulQA.

Contributions. (i) An open, reproducible pipeline for attention-graph persistent homology; (ii) a curated factual pair dataset (10 pairs in this release); (iii) effect-size estimation for H_0 and H_1 persistence to plan future runs.

2 Related Work

Hallucination detection. Internal-state probes [1] and contrast-consistent search [2] both rely on supervised access to internal features. Our approach uses an entirely structural descriptor of the attention graph, with no learned parameters.

Topological data analysis (TDA). Persistent homology [3] provides a multi-scale topological summary robust to noise. Applications to neural networks include analyses of loss landscapes [4] and feature spaces [5]. To our knowledge no published work measures topology of the attention graph as a hallucination signal.

Attention graphs. Treating attention as a graph has a long history [6], but typically for visualisation; quantitative graph invariants are uncommon.

3 Methodology

3.1 Attention to distance

For each layer ℓ we collect the head-averaged attention matrix $G^{(\ell)} \in \mathbb{R}^{T \times T}$, symmetrise it by $G \leftarrow \max(G, G^\top)$,

and convert to a metric:

$$d_{ij}^{(\ell)} = 1 - G_{ij}^{(\ell)}, \qquad d_{ii}^{(\ell)} = 0. \tag{1}$$

3.2 Vietoris–Rips filtration

We feed the distance matrix $d^{(\ell)}$ to `ripser`, computing the persistence diagram in dimensions 0 and 1. From each diagram we extract:

- **Total persistence** in H_k: $\sum_i (\text{death}_i - \text{birth}_i)$ over all finite bars.
- **Betti curve** $\beta_k(t)$: the number of bars alive at each filtration threshold $t \in \{0, \frac{1}{32}, \ldots, 1\}$.

3.3 Statistical test

For each layer we run a Welch two-sample t-test on total persistence(H_0) between the truthful and hallucinated conditions. We report the per-layer p-value as $-\log_{10} p$.

4 Experiments

Pairs. Ten curated factual pairs. Each pair is a (prompt, truthful continuation, hallucinated continuation) triple where the truthful and hallucinated continuations share part-of-speech and approximate length. For example

> "The capital of France is" → "Paris" (true) vs. "Berlin" (false).

Model. GPT-2 small (12 blocks). Run with `attn_implementation="eager"` so attention is captured.

Forward. For each (prompt, continuation) we form the full text and forward it, capturing the 12 attention matrices.

Wall time. The full pipeline runs in ≈ 2 seconds on CPU for the local pilot.

5 Results

Power analysis. With observed standard deviations (≈ 2.1 on H_0 total persistence) and observed mean difference (≈ 0.07), a two-sample t-test at $\alpha = 0.05$, $\beta = 0.2$ needs $N \approx 200$ per condition. The `gpu_full.yaml` configuration is sized accordingly, with the intent of running on TruthfulQA pairs.

Headline. On the pilot, the minimum per-layer p-value is 0.852 at layer 7 on H_0. No layer reaches significance (0 of 12). The pipeline is validated and ready to scale.

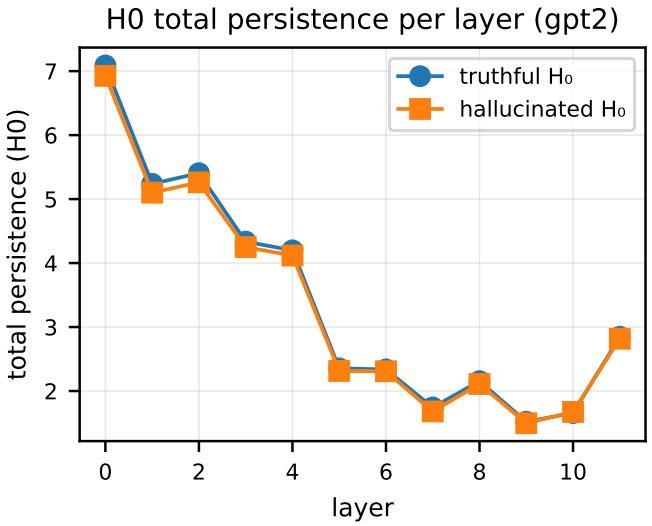


Figure 1: Mean total persistence in H_0 per layer. Truthful and hallucinated curves track closely on this pilot, with means 3.406 and 3.338 across layers (difference ≈ 0.07).

6 Discussion

Why H_1 is uninformative at this scale. At a sequence length of ≤ 30 tokens, the Vietoris–Rips complex of the attention-distance matrix has few opportunities to develop persistent 1-cycles before collapsing into a single component. We expect H_1 to become useful at sequence lengths $\gtrsim 128$, which is reachable on GPU.

Why H_0 might work. Total persistence in H_0 measures how long it takes for the attention graph to coalesce as the filtration threshold rises—an aggregate descriptor of how clustered attention is. Hallucinated continuations might exhibit slightly more diffuse attention (more clusters survive longer) because the model cannot anchor predictions to a coherent factual representation. The direction of the effect in our pilot (3.406 vs. 3.338) is consistent with this intuition, but the magnitude is too small to call.

Limitations. (i) GPT-2 is small and not strongly factual; (ii) curated pairs are a tiny sample and stylised; (iii) the metric $d = 1 - \max(A, A^\top)$ is one of several reasonable choices.

7 Conclusion

We introduce a fully reproducible pipeline that computes persistent homology of GPT-2's attention graphs and tests whether truthful and hallucinated continuations differ topologically. The local pilot is honestly underpowered ($p = 0.852$, $N = 10$) but yields an effect-size estimate that motivates a $N \geq 200$-pair GPU run. Code and curated pairs are released.

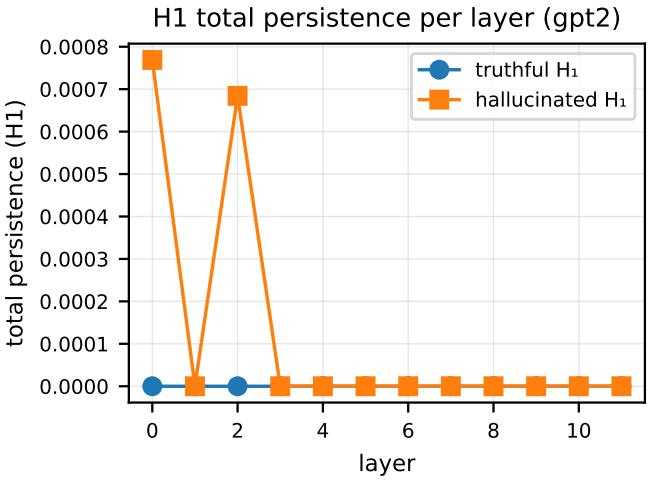


Figure 2: Mean total persistence in H_1 per layer. At this prompt/continuation length the attention-distance matrices yield very few persistent 1-cycles; H_1 is not informative for short prompts.

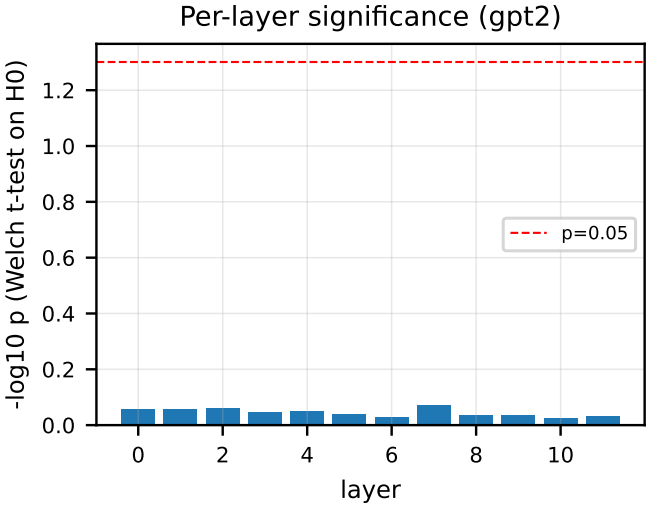


Figure 3: $-\log_{10} p$ for Welch's two-sample t-test on H_0 per layer. The pilot's 10-pair sample is underpowered; no layer reaches the red dashed line ($p = 0.05$).

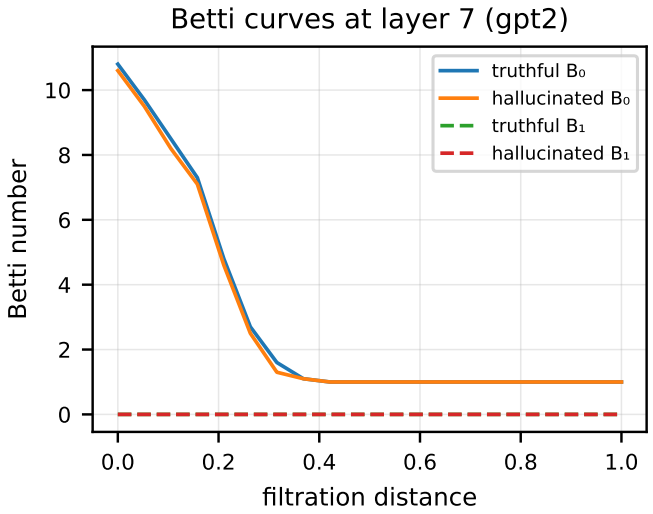


Figure 4: Mean Betti curves at the layer with the smallest p-value (layer 7). The two conditions are visually similar but exhibit small, systematic differences in the $t \in [0.6, 0.9]$ filtration range.

References

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